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DUNMAN HIGH SCHOOL

Preliminary Examination

Year 6

MATHEMATICS (Higher 1)

8864

Paper 1

26 September 2013

3 hours

Additional Materials: Answer Paper
List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your Name, Index Number and Class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer all the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, **attach the question paper to the front of your answer script.**

The total number of marks for this paper is **95**.

For teachers' use:

Qn	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Total
Score													
Max Score	5	6	7	7	10	6	7	8	8	10	10	11	95

Section A: Pure Mathematics [35 marks]

- 1 Sketch the graphs of $y = \frac{2}{(x+1)^2}$ and $y = x^2 - 3$ on the same diagram, showing clearly the asymptotes, the axial intercepts and the stationary point. [2]
Hence solve the inequality $x^2 \leq \frac{2}{(x+1)^2} + 3$. [3]
- 2 (a) If $a \neq b$, show that the equation $\frac{1}{3}x^2 + ax + 3ab = -bx$ has two distinct real roots for all real, non-zero values of a and b . [3]
(b) Without the use of the graphic calculator, solve $(1-3x)^2 \leq 4$. [3]
- 3 (a) Find $\int \frac{(1-\sqrt{x})^2}{2\sqrt{x}} dx$. [4]
(b) Find the exact area enclosed by the curve $y^3 = x-1$, the x -axis and the lines $x=2$ and $x=9$. [3]
- 4 Without the use of a calculator, solve
(a) $3e^x = 5 + 2e^{-x}$, [3]
(b) $2 + 2\log_4(x-1) = \log_2 x$. [4]
- 5 (a) A curve has equation $y = \ln x - e^{2x} - x$. Using differentiation, find the coordinates of the stationary point and determine its nature. [4]
Given that the tangent at $x=1$ passes through the point $(-1, a)$, where a is a real constant, determine the exact value of a . [3]
(b) The surface area of a sphere is increasing at a constant rate of $7 \text{ cm}^2 \text{ s}^{-1}$. Find the exact rate of change of the radius at the instant when its volume is $4.5\pi \text{ cm}^3$. [3]
[The surface area of a sphere of radius r is $4\pi r^2$ and the volume is $\frac{4}{3}\pi r^3$.]

[Turn over]

Section B: Statistics [60 marks]

- 6** The lengths, in cm, of fish in a large pond are normally distributed with mean 25 cm and variance σ^2 . It is assumed that the fish population is large.

- (i)** Given that 85% of the fish are longer than 20 cm, show that the value of σ is 4.82, correct to 3 significant figures. [2]
- (ii)** The price of a fish is $0.1x$ dollars where x is the length of the fish in cm. Find the expected total price of 100 fish. [1]

A fish is ready for the market if it is more than 28 cm.

- (iii)** Find the probability that a fish randomly taken out of the pond is ready for the market. [1]
- (iv)** Three fish are randomly taken out of the pond. Find the probability that exactly two fish are ready for the market. [2]

- 7** A lemonade dispensing machine dispenses S ml of syrup and W ml of water in each cup of lemonade. S and W are independent random variables which follow the distributions $N(50, 3^2)$ and $N(140, 5^2)$ respectively.

- (i)** Find the probability that the volume of water exceeds three times the volume of syrup by more than 5 ml in a randomly chosen cup of lemonade. [2]
- (ii)** The machine initially contains 10 litres of syrup and an unlimited supply of water. If the probability that the machine is able to dispense n cups of lemonade exceeds 0.99, find the greatest value of n . (Answers obtained by trial and improvement from a calculator will obtain no marks.) [5]

[Turn over]

- 8** At a car-wash service centre, the random variable X is the time taken for a new team of workers to wash a car. X has a mean 8.6 minutes and standard deviation 1.5 minutes.

(i) The mean washing time of a random sample of 10 cars is denoted by $\bar{X}$. State the values of $E(\bar{X})$ and $Var(\bar{X})$. [2]

After six months, the team is more experienced.

(ii) Find the new mean time to wash a car, if the probability that the total washing time for a random sample of 60 cars is now less than 402 minutes is 0.6. (You may assume that the standard deviation for washing a car remains unchanged.) [6]

- 9** A box contains 5 red balls, 5 green balls and 5 blue balls. The balls are identical in size. Balls of the same colour are each printed with a distinct number from 1, 2, 3, 4, 5. Two balls are randomly drawn without replacement. Find the probability that

(i) at least 1 red ball is drawn, [2]

(ii) the sum of the numbers on the balls drawn is 9, [2]

(iii) the sum of the numbers on the balls drawn is 9 given that at least 1 red ball is drawn. [3]

State, with a reason, whether the events ‘the sum of the numbers on the balls drawn is 9’ and ‘at least 1 red ball is drawn’ are independent. [1]

[Turn over]

- 10 (a)** In a large company, the percentage of directors, managers and executives is 5%, 20% and 75% respectively. A random sample of 60 staff members is chosen to take part in a survey to determine the productivity of staff in the company. Describe how a stratified sample can be obtained and state an advantage of this method of sampling. [3]

- (b)** One question in the survey is as follows:

“Do you use Facebook during office hours at least once a day? Yes or No”

It is known that on average, 7 out of 9 staff members answered “yes” to the above question.

For the random sample of 60 staff members selected in **(a)**,

- (i)** find the standard deviation of the number of staff members who answer “yes”, [1]
- (ii)** find the probability that the number of staff members who answer “yes” exceeds the mean, [3]
- (iii)** find the most probable number of staff members who answer “yes”, [2]
- (iv)** state, with a reason, if the above answers are valid if the total number of staff members in the company is small. [1]

- 11 (a)** Jerome claims that customers have to queue at a supermarket check-out counter for more than 5 minutes, on average.

To investigate the claim, the queuing times, x minutes, of 50 randomly selected customers are obtained and summarised below:

$$\sum x = 270, \quad \sum (x - \bar{x})^2 = 64.19.$$

- (i)** Find the unbiased estimates of the population mean and variance. [2]
 - (ii)** Test, at the 1% significance level, whether Jerome’s claim is valid. [4]
 - (iii)** State, giving a reason, whether it is necessary to assume a normal distribution for the test to be valid. [1]
- (b)** The same sample is used to test the null hypothesis $\mu = \mu_0$ against the alternative hypothesis $\mu \neq \mu_0$ at 1% significance level, where μ denotes the population mean queuing time. Find the set of possible values of μ_0 if the null hypothesis is rejected. [3]

[Turn over]

- 12** At a certain restaurant, the time taken between a customer sitting down at a table and his order being taken by a waiter is dependent on the number of other customers already seated in the restaurant.

Carl is a customer who visited the restaurant on eight different occasions. The table below shows the number, x , of customers already seated and the time taken, y in minutes, for a waiter to take Carl's order.

x	3	8	9	10	7	12	2	11
y	6	9	11	11	9	13	4	12

- (i)** Plot a scatter diagram of these data. [2]
- (ii)** Write down the product moment correlation coefficient and comment on the relationship between x and y . [2]
- (iii)** Calculate the equation of the regression line of y on x in the form $y = a + bx$. [1]
- (iv)** Give an interpretation, in context, of the values obtained for a and b . [2]
- (v)** Use the equation of the regression line in **(iii)** to estimate the time taken for a waiter to take Carl's order when the number of customers already seated in the restaurant is
 - (a)** 5,
 - (b)** 20. [2]
- (vi)** Giving reasons, comment on the reliability of each of your estimates obtained in **(v)**. [2]

END OF PAPER